

Receive a Customer Return SOP

One episode receives one customer return. It runs from one closed return mailer standing at the back of the table, a bare card spot, a bare inspect mat, an empty paperwork tray and three empty grade bins, to an open empty mailer, one labelled item lying in the grade bin that matches its letter, and the RMA card lying flat in the paperwork tray.

The order never changes: open the mailer, take out the card and the item, verify the contents against the card, grade the item A, B or C, then route the item and the paperwork. Nothing is graded before it is verified, and nothing is routed before it is graded and labelled. The item is routed first and the paperwork second.

The right gripper does the primary work: it turns the flap back, lifts the item out, turns the item for the look, peels and sticks the grade label, and carries the item to its bin. The left gripper supports: it holds the mailer down while the item comes out, holds the item still while the label goes on, and it owns the paperwork, lifting the RMA card out and filing it. The mailer is always held down by the arm that is not lifting, so the right gripper holds it while the card comes out. The one two armed action is the open: both grippers free their own tab together, each on its own end, before the right gripper turns the flap back. The right arm is the working arm because the rig is set up right handed, so the dominant hand drives it in leader follower. A left handed collection requires a separately approved mirrored variant. Do not swap arms during an episode.

Neither gripper reaches onto the other arm's side to pick anything up or to carry it. One object sits on the midline and is shared: the return mailer, which each gripper meets from its own side, the right from the right and the left from the left. The inspect mat straddles the midline and both arms work over it. Nothing is ever passed from one gripper to the other, only one object is held at a time, and no grasp is changed while an object is off the surface.

The counts are fixed: one return, one RMA card, one item, one grade, one label, one bin. There is no second grade and no second label, and a grade is never changed once the label is on. If the item and the card do not match, the item is graded C and the episode continues. The mailer stays on the table for the whole episode and is never lifted, tilted or turned over.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 90 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the return spot, the label sheet spot, the card spot, the inspect mat, the paperwork tray and all three grade bins.
3. The printed side of the RMA card and every face of the item stay in frame for the whole check, and the letter on the grade label is readable before the item goes into a bin.
4. Both arms are at home with their grippers open.
5. The tabletop is dry and clear of anything but the items in the Materials checklist.
6. The right arm reaches the label sheet spot, the mailer from its right side, the inspect mat and all three grade bins without extending to a joint limit or folding in on itself. Bring the bins forward until this is true.
7. The left arm reaches the mailer from its left side, the card spot, the inspect mat and the paperwork tray without extending to a joint limit.
8. Neither arm needs the far side of the table for anything. The paths from the mailer forward to the mat, from the mat right to the bins, and from the card spot forward to the tray are unobstructed. Carry the item forward by hand during setup: it must clear the standing mailer on both sides. Widen the gap between the mailer and the mat until it does.

Materials checklist

1. One return mailer stands closed at the return spot at the back center of the table, straddling the midline, its flap up and its shipping label facing up.
2. The mailer is a stiff box mailer with one fold over flap held down by one tab at each end. It is held shut by the tabs and by nothing else. A mailer closed with tape, glue or staples is not usable: nothing on this station cuts.
3. The mailer is low enough that the gripper reaches the contents from above without passing below the mailer's top edge, and wide enough that the card and the item lift straight out without touching a wall. It is no taller than it is stable: a mailer a passing item could brush over is replaced or brought forward.
4. The mailer holds one RMA card lying face up on top of one item, and nothing else. No loose fill, no second item, no second card.
5. The RMA card is printed on one side and carries the RMA number, the part number, the item name and the line count. The line count is one.
6. The item is one single unit, dry, light enough for one gripper to carry level, and square enough for the gripper to close on it and hold it level. It has one flat clear face large enough to take a grade label, away from its own part number label.
7. The card spot, left of the midline, is bare.
8. The inspect mat lies flat at the front center, straddling the midline, and is bare. It grips the table: draw a hand across it during setup, and if it slides, bunches or lifts, weight it or

replace it with a rubber mat that stays put. Nothing pins the mat during the episode.

9. One label sheet lies flat on the label sheet spot, back right, on the right arm's side. It carries at least one A label, one B label and one C label. Each letter is large enough to read in the environment camera, and each label peels off in one piece.
10. Three grade bins stand in a row at the front right, on the right arm's side, marked A, B and C from left to right, with a clear gap between them. All three are empty, shallow enough that the item is visible inside, and wide enough that the item goes in without being pushed.
11. The paperwork tray stands at the front left, on the left arm's side, and is empty. It is shallow enough that the card lies flat inside it and does not stand on the rim.

Workspace layout

Three rows: the mailer and the label sheet at the back, the card spot and the inspect mat in the middle, the tray and the bins at the front. Everything travels forward, then the paperwork goes left and the item goes right.

Back row, left to right:

1. Return spot, back center on the midline, the mailer standing closed with its flap up. Shared: the right gripper works it, the left gripper holds it, and each meets it from its own side.
2. Label sheet spot, back right, on the right arm's side. Nothing stands at the back left.

Middle row, left to right:

3. Card spot, left of the midline, in front of the mailer. Bare at the start, the RMA card lies there face up from Step 2 until Step 5.
4. Inspect mat, front center, straddling the midline. Bare at the start, the item lies there from Step 2 until Step 5.

Front row, left to right:

5. Paperwork tray, front left, on the left arm's side. Empty at the start, one RMA card lying flat in it at the end.
6. Grade bins, front right, on the right arm's side, marked A, B and C from left to right. All empty at the start, one item in one of them at the end.

A clear gap of at least a gripper's width sits between the mailer and the inspect mat and between the mailer and the card spot, so an object lifts straight up out of the mailer and travels forward without touching it. No carry passes over another bin, over the tray or over the mailer, and no gripper travels to the far side of the table.

Vocabulary

- **Arm assignment:** the right gripper turns the flap back, lifts the item out, turns the item for the look, sticks the label and carries the item to its bin. The left gripper holds the mailer,

holds the item, and carries and files the RMA card. Both grippers free their own tab together in Step 1, each on its own end. Neither reaches onto the other's side to take or carry anything.

- **Hold:** a closed empty gripper pressed onto an object to keep it still. A hold never lifts, carries or moves what it holds.
- **Home spot:** the spot an object is laid on and lies flat on before the next step starts: the card spot, the inspect mat, the paperwork tray, a grade bin. Nothing is left in the mailer, on bare tabletop or in a gripper between steps.
- **Release point:** the gripper opens over the card spot, over the inspect mat, over the paperwork tray, or inside the mouth of a grade bin. Never over the mailer, never over another bin and never over open tabletop.
- **Stable placement:** the object lies still after release, flat on its surface, and does not rock, tip, slide or lean on anything.
- **Return mailer:** the closed box the return arrives in. It stands at the return spot and is worked in place.
- **Flap:** the fold over lid of the mailer, held down by one tab at each end.
- **Tab:** one of the two card tongues that hold the flap down, one at the left end and one at the right end. Each gripper frees the tab on its own side.
- **Open mailer:** the flap turned back flat against the back of the mailer, both tabs out of their slots, and the contents visible from above.
- **RMA card:** the printed card that came in the mailer. It says what the return should be.
- **RMA number:** the number printed on the card. It must be the same as the number on the mailer's shipping label.
- **Part number:** the number printed on the card and on the item that says which item it is.
- **Line count:** the number printed on the card that says how many items the mailer should hold. It is one in every episode.
- **Item:** the single returned unit that came in the mailer. It is the thing that is graded.
- **Inspect mat:** the flat mat the item lies on while it is checked and graded.
- **Face:** one side of the item. Every face is brought into the camera frame in Step 4.
- **Face look:** one face brought into the camera frame and held still for 1 second. The item is kept low over the mat and is set back down between faces.
- **Mark:** anything on the item that was not there when it left: a scuff, a scratch, a dent, a tear, a stain, a peeling label, or a missing bag.
- **Match:** all three of these are the same: the part number on the item and the part number on the card, the RMA number on the card and the number on the mailer's shipping label, and the count of items in the mailer and the line count.
- **Mismatch:** any one of those three is not the same. A mismatch is graded C.

- **Grade A:** the item is clean and whole and there is no mark on any face. It can go back on the shelf as it is.
- **Grade B:** the item is whole and works, but there is at least one mark you can see. It needs work before it sells again.
- **Grade C:** the item is broken, cracked, wet or missing a part, or the contents are a mismatch. It does not go back on the shelf.
- **Lower grade wins:** when the item could be either of two grades, take the lower one. A or B becomes B, and B or C becomes C.
- **Grade label:** the sticker printed A, B or C that is stuck on the item to show its grade.
- **Labelled item:** an item with exactly one grade label stuck flat on one face, fully on the item, readable from above, and clear of the item's own part number label.
- **Grade bin:** one of the three bins at the front right. The item goes in the bin whose letter is the same as the letter on its label.
- **Filed card:** the RMA card lying flat and printed side up inside the paperwork tray.
- **Quarter turn:** the most any object is tilted in a gripper during a carry. Turning the item for the look is done low over the mat, and nothing is ever turned over in the air.

Steps

Step 1: Open the return mailer

Goal: both tabs are out of their slots, the flap lies back flat, and the RMA card is visible face up on top of the item.

The mailer stays at the return spot for the whole episode. It is never lifted, tilted or turned over.

1.1 Free the two tabs

- Bring both grippers down on the flap, the left gripper on the left tab and the right gripper on the right tab.
- Lift each tab straight up out of its slot together, each gripper on its own end, keeping the flap level.
- Release both tabs and lift clear.

The flap is held at both ends. One arm lifting one end leaves the other end hooked, so the flap bends and the corner tears. Both arms free their own tab first, so the flap comes up level.

1.2 Turn the flap back

- With the left gripper closed and empty, come down on the left side of the mailer's top edge and press it against the table.
- With the right gripper, take the front edge of the flap, lift it, and turn it back against the back of the mailer until it lies flat and clear of the mouth.

- Release the flap and lift clear.
- The left gripper releases the mailer and lifts clear only now.

Never force the flap. If a tab is still hooked, lay the flap down, free that tab again, then turn the flap back.

Check: the flap lies back flat, both tabs are out of their slots, nothing is torn, the mailer still stands at the return spot, and the RMA card is visible face up on top of the item. If the flap springs back over the mouth, hold it back with the right gripper and turn it again. Make no more than one correction. If a tab or a corner is torn, end the episode and log a defective mailer.

Step 2: Take out the card and the item

Goal: the RMA card lies face up on the card spot, the item lies on the inspect mat, and the mailer is empty.

The card lies on top, so the card comes out first and nothing is ever lifted over it. One object per carry. Each object is lifted straight up until it is clear of the mailer's top edge before any horizontal motion, and nothing is dragged up a mailer wall.

2.1 Take out the RMA card

- With the right gripper closed and empty, come down on the right side of the mailer's top edge and press it against the table. Take this hold before the left gripper enters the mailer.
- With the left gripper, close on the near edge of the RMA card and lift it straight up until it is clear of the mailer's top edge.
- Carry it forward and left to the card spot, lay it flat printed side up, release, and lift clear.
- The right gripper releases the mailer and lifts clear only now.

Check: the card lies flat and printed side up, fully on the card spot, and is not bent, torn or upside down. If it lands face down or off the spot, the left gripper lifts it clear and lays it again. Make no more than one correction.

2.2 Take out the item

- With the left gripper closed and empty, come down on the left side of the mailer's top edge and press it against the table. Take this hold before the right gripper enters the mailer.
- With the right gripper, close on the item square, so it hangs level and does not lean in the gripper.
- Lift it straight up until it is clear of the mailer's top edge, carry it forward level, and lower it onto the middle of the inspect mat.
- Release once it lies on its own, and lift clear.
- The left gripper releases the mailer and lifts clear only now.

The item is never dragged up a mailer wall, and the mailer is never leaned over to tip the item out.

Check: the item lies still on the inspect mat, fully on it, and the mailer is empty. If a second item, a second card or loose fill is still in the mailer, end the episode and log a reset error: the count is fixed at one of each.

Step 3: Verify the contents against the card

Goal: the part number, the RMA number and the count have all been checked against the card, and the result is either a match or a mismatch.

Nothing is graded and nothing is moved to a bin in this step. Both grippers stay clear of the card and the item so both stay in the camera frame.

3.1 Read the card

- Bring both grippers clear of the card spot and of the inspect mat.
- Hold still for 2 seconds with the printed side of the card in frame, so the RMA number, the part number and the line count are all readable.

3.2 Check the three lines

- **Part number:** the number on the item is the same as the part number on the card.
- **RMA number:** the number on the card is the same as the number on the mailer's shipping label.
- **Count:** the mailer held one item, the same as the line count on the card.
- If the item's part number is not readable from above, the right gripper turns the item on the mat until it is, then sets it down again.

All three the same is a match. Any one of them not the same is a mismatch.

Check: all three lines have been checked and the result is one of match or mismatch. A mismatch is not put right on the table. Nothing is swapped, nothing is written on the card, and nothing is sent back to the mailer. The mismatch is carried into Step 4 as a grade C.

Step 4: Grade the item A, B or C

Goal: every face of the item has been looked at, one grade has been chosen, and one grade label is stuck flat on the item.

The grade is chosen once. It is never changed after the label is on, and only one label is ever used.

4.1 Look at every face

- With the right gripper, take the item, tilt it no more than a quarter turn to bring one face into the camera frame, hold it still for 1 second, then set it back down on the mat.

- Work around the item until every face has been brought into the frame and held.
- Keep the item low over the mat the whole time. Do not turn it over in the air and do not carry it off the mat.

4.2 Choose the grade

- **A** if the item is clean and whole and there is no mark on any face.
- **B** if the item is whole and works, but there is at least one mark you can see on any face.
- **C** if the item is broken, cracked, wet or missing a part, or the contents are a mismatch from Step 3.
- If the item could be either of two grades, take the lower one. The lower grade always wins.

4.3 Stick one grade label on

- With the left gripper closed and empty, come down on the item and hold it still on the mat.
- With the right gripper, peel one label off the label sheet: the A label for grade A, the B label for grade B, the C label for grade C.
- Bring the label over a flat clear face of the item, lower it, and press it down flat.
- Release and lift clear.
- The left gripper releases the item and lifts clear only now.

One label, and one only. A label that lands crooked is left as it is. It is not peeled off again and it is not covered with a second label.

Check: exactly one grade label is stuck flat on the item, fully on one face, readable from above, and clear of the item's own part number label. The item still lies on the inspect mat and the label sheet still lies flat on its spot. If a corner of the label is hanging off, press it down once with the right gripper. Make no more than one correction.

Step 5: Route the item and the paperwork

Goal: the item lies in the grade bin whose letter is the same as the letter on its label, the RMA card lies flat in the paperwork tray, and the inspect mat and the card spot are both bare.

The item goes first and the paperwork second. The card is never filed before the item is in its bin.

5.1 Put the item in its grade bin

- With the right gripper, close on the item square, lift it straight up off the mat, and carry it level to the right.
- Read the letter on the grade label and go to the bin marked with the same letter.
- Bring the item inside the mouth of that bin and low to the bottom, then release.
- Lift clear of the bin.

The item never passes over another bin on its way to its own, and it is never dropped in from above the bin's rim.

5.2 File the RMA card

- With the left gripper, close on the near edge of the RMA card and lift it straight up off the card spot.
- Carry it forward to the paperwork tray, keeping it level.
- Lay it flat printed side up inside the tray, release, and lift clear.

The card never passes over the inspect mat and never over a grade bin.

Check: the item lies in one bin and one bin only, the letter on its label is the same as the letter on that bin, and it lies flat inside. The card lies flat and printed side up in the tray. The inspect mat and the card spot are both bare and the other two bins are still empty. If the item has gone into the wrong bin, the right gripper lifts it clear and places it in the right one. Make no more than one correction. If the card sits on the rim of the tray, the left gripper lifts it clear and lays it in again, once.

Step 6: End the episode

- Confirm the mailer: empty, standing at the return spot, the flap back and both tabs out.
- Confirm the item: in the bin that matches its label, lying flat, with one label on it.
- Confirm the paperwork: one RMA card lying flat and printed side up in the tray.
- Confirm the table: the card spot bare, the inspect mat bare, the label sheet flat on its spot, and the other two bins empty.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return both arms home with their grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the item out of its bin, peel the grade label off, and wipe the item dry. Check that the run has not damaged it.
2. Take the RMA card out of the paperwork tray and keep it with its own item.
3. Stand the item back in the mailer, then lay the card face up on top of it. One item and one card, nothing else.
4. Turn the flap forward over the mouth and push both tabs back into their slots. Replace a mailer whose tabs no longer hold or whose corners are torn.
5. Stand the mailer back on the return spot, straddling the midline, flap up and shipping label facing up.

6. Empty all three grade bins and stand them back in a row at the front right, A, B and C from left to right, with a clear gap between them. Empty the paperwork tray.
7. Wipe the inspect mat and the tabletop and dry both. Confirm the mat lies flat and still grips the table.
8. Lay a fresh label sheet flat on the label sheet spot whenever fewer than one A, one B and one C label are left on it.
9. Vary the return between episodes. Change the item, change the marks on it and change the numbers on the card, so that all three grades and both a match and a mismatch are collected.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** the item comes out before the card, a grade label is taken or stuck before the three lines in Step 3 have been checked, the item leaves the mat for a bin before its label is on, the card is filed before the item is in its bin, or a step is skipped.
- **SOP rule broken:** Steps 1 to 5, open, take out, verify, grade, route, in that order, card out first and item routed before paperwork.
- **Coaching note:** open it, empty it, check it, grade it, route it. The card comes out first because it lies on top, and it is filed last because the item leads.

Violation: Arms swapped roles

- **Visible cue:** the left gripper turns the flap back, lifts the item out, peels or sticks the label, or carries the item to a bin; the right gripper lifts or carries the RMA card or reaches the paperwork tray; or either gripper reaches onto the other arm's side for an object that belongs to that arm. The two armed tab lift in Step 1 is not a swap: there each gripper works its own end of the flap.

- **SOP rule broken:** Steps 1 to 5, the right gripper opens, lifts the item, labels it and bins it, and the left gripper holds and owns the paperwork, with both arms freeing their own tab together in Step 1.
- **Coaching note:** if an arm is reaching across, the wrong arm has been chosen. Right owns the item, left owns the paper.

Violation: Mailer not held, lifted, or knocked over

- **Visible cue:** a gripper enters the mailer while the other gripper is off it; the mailer lifts, shifts or tips as the card or the item is drawn out; or the mailer leaves the table, falls on its side, or is knocked off the return spot.
- **SOP rule broken:** Steps 1.2, 2.1 and 2.2, the mailer is held down by the arm that is not lifting, and it stays standing on the table throughout.
- **Coaching note:** the holding arm takes its hold first and lets go last. Nothing in this task requires picking the mailer up.

Violation: Flap forced or torn

- **Visible cue:** one gripper lifts one end of the flap while the other tab is still in its slot, the flap is pulled while a tab is hooked, a tab or a corner tears, or the flap is left standing over the mouth instead of being turned back flat.
- **SOP rule broken:** Steps 1.1 and 1.2, both grippers free their own tab together, then the flap is turned back flat against the back of the mailer.
- **Coaching note:** free both tabs, then turn it back. A flap that will not turn back has a tab still hooked, so free it again rather than pulling harder.

Violation: Contents dragged out or tipped out

- **Visible cue:** the card or the item is drawn up a mailer wall instead of lifted straight up, moves sideways before it is clear of the top edge, is scraped over the rim, or the mailer is leaned over so the contents fall out.
- **SOP rule broken:** Step 2, each object is lifted straight up until it is clear of the mailer's top edge before any horizontal motion, and the mailer is never tilted.
- **Coaching note:** straight up first, then across. The rim is where returns get scratched.

Violation: Card mishandled

- **Visible cue:** the card ends face down, off the card spot, bent, folded, torn, or left in the mailer; the card is laid on the inspect mat, on bare tabletop or on the item; or the card is still in a gripper when Step 3 starts.
- **SOP rule broken:** Step 2.1, the card lies flat and printed side up, fully on the card spot, before the item comes out.
- **Coaching note:** the card is the record. Printed side up, flat, and on its own spot every time.

Violation: Verify skipped or shortened

- **Visible cue:** a grade is chosen with no pause over the card, the card is never held still and readable in frame, or fewer than the three lines are checked: part number, RMA number and count.
- **SOP rule broken:** Step 3, the card is read with both grippers clear for 2 seconds, and all three lines are checked.
- **Coaching note:** the check is a step, not a glance. Three lines, every time, in the same order.

Violation: Card or item out of frame during the check

- **Visible cue:** a gripper covers the printed side of the card or the item's part number during Step 3, or the card or the item is moved out of the environment camera frame while the check runs.
- **SOP rule broken:** Step 3, both grippers stay clear so the card and the item both stay in frame.
- **Coaching note:** hands off during the read. If the reviewer cannot see it, the check did not happen.

Violation: Mismatch handled on the table

- **Visible cue:** after a mismatch, the item is put back in the mailer, swapped for another item, set aside off its spots, or something is written on or added to the card; or the episode is stopped instead of run to the end.
- **SOP rule broken:** Steps 3 and 4.2, a mismatch is graded C and the episode continues.
- **Coaching note:** a mismatch is a grade, not a repair. Label it C, bin it C, file the card, and let review pick it up.

Violation: Faces not all looked at

- **Visible cue:** a grade is chosen before every face has been brought into the frame, a face is turned past the frame without being held still, or the hold on a face is clearly shorter than 1 second.
- **SOP rule broken:** Step 4.1, every face is brought into the camera frame and held still for 1 second.
- **Coaching note:** the mark is usually on the face you did not turn up. Every face, held still, every time.

Violation: Item turned over in the air or carried off the mat

- **Visible cue:** during the look, the item is tilted past a quarter turn, rolled over in the gripper, held high above the mat, or carried away from the inspect mat.
- **SOP rule broken:** Step 4.1, the item is kept low over the mat, tilted no more than a quarter turn, and set back down between faces.

- **Coaching note:** low and slow, and down between faces. A dropped return becomes a grade C that the customer did not cause.

Violation: Wrong grade for what is visible

- **Visible cue:** an item with a visible mark is graded A, a broken, cracked, wet or incomplete item is graded A or B, a mismatch is graded A or B, or an item that could be either of two grades is given the higher one.
- **SOP rule broken:** Step 4.2, A is clean and whole, B is whole with a mark, C is broken or incomplete or a mismatch, and the lower grade always wins.
- **Coaching note:** when it is between two grades, take the lower one. A wrong A costs more than a wrong B.

Violation: Grade changed after the label is on

- **Visible cue:** a label is peeled back off the item, a second label is stuck over the first, or the item is put in a bin whose letter is not the letter on its label because the grade was reconsidered.
- **SOP rule broken:** Step 4, one grade and one label, and the grade is never changed once the label is on.
- **Coaching note:** decide before the sticker. Once it is on, it is the answer.

Violation: Label count wrong or label placed wrong

- **Visible cue:** no label is stuck on, more than one label is stuck on, the label lands half off the item, folds over an edge, sits on the mat, covers the item's own part number label, or is not readable from above.
- **SOP rule broken:** Step 4.3, exactly one grade label, stuck flat on one clear face, fully on the item, readable from above and clear of the part number.
- **Coaching note:** one label, flat, on a clear face. A crooked label that is stuck down stays: peeling it off is the bigger error.

Violation: Item routed to the wrong bin

- **Visible cue:** the item ends in a bin whose letter is not the letter on its label, in more than one bin across the episode with no correction called, or on the tabletop beside the bins.
- **SOP rule broken:** Step 5.1, the item goes in the bin whose letter is the same as the letter on its label.
- **Coaching note:** read the label at the bin, not from memory at the mat.

Violation: Item dropped into the bin

- **Visible cue:** the gripper opens above the bin's rim, the item falls a visible distance, or the item bounces or lands on its edge inside the bin.

- **SOP rule broken:** Step 5.1, the item is brought inside the mouth of the bin and low to the bottom before the gripper opens.
- **Coaching note:** reach in, then let go. The bin is not a target to throw at.

Violation: Paperwork not filed, or filed early

- **Visible cue:** the card is carried to the tray before the item is in its bin, the card ends face down or on the rim of the tray, the card is left on the card spot at the end, or the card is put into a grade bin.
- **SOP rule broken:** Step 5.2, the card is filed after the item is binned, and it lies flat and printed side up inside the tray.
- **Coaching note:** item first, paper second. The card and the item are one record, so neither ends without the other.

Violation: A carry passes over another station

- **Visible cue:** the item passes over another grade bin, over the paperwork tray or over the mailer; or the card passes over the inspect mat or over a grade bin.
- **SOP rule broken:** Steps 2, 5.1 and 5.2, each object travels on a clear path to its own spot and over nothing else.
- **Coaching note:** route around, not across. What falls mid carry lands in the wrong bin.

Violation: Bad pick, carry, or release

- **Visible cue:** an object moves from one gripper to the other without being set down first, both grippers hold an object at once, a gripper shifts or seats its hold again while the object is off the surface, the item or the card slides along the table with no daylight under it, the item hangs tilted in the gripper because the grasp was taken off square, or an object is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 5, one object at a time, lifted straight up clear of the surface before any horizontal motion, gripped square, tilted no more than a quarter turn, and never passed between grippers.
- **Coaching note:** set it down to change the grip, and lift before the carry. An item that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** the item, the card, a bin, the tray or the mat is shoved, nudged or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 5, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, a gripper or a carried object strikes the mailer, a bin, the tray, the mat or the label sheet and knocks, tips, slides or displaces it; the two arms strike each other over the mailer; or an object falls short of its spot, with no gripper failure, drift or controller fault visible.
- **SOP rule broken:** Steps 1 to 5, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. On the mailer, the working arm moves and the holding arm stays put.

Violation: Manipulation outside the camera frame

- **Visible cue:** a tab lift, a take out, a face look, a label press, a bin placement or the filing of the card happens partly or wholly outside the environment camera frame.
- **SOP rule broken:** Steps 1 to 5, every manipulation happens inside the camera frame.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the item still on the mat or still in the mailer, the card still on the card spot, an unlabelled item in a bin, something left in a gripper, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 6, confirm the mailer, the item, the paperwork and the table, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside an arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective mailer: one whose tabs will not hold the flap down, whose flap will not stay back once it is turned, or whose corners are already torn at episode start.
- Defective mat: one that slides, bunches or lifts under a correct carry, or that will not lie flat. Weight or replace it before the next episode.

- Defective label sheet: one whose labels tear or split when peeled, or that has fewer than one A, one B and one C label left.
- Defective bin: one too deep to see the item in, too narrow for the item to go in without being pushed, or that will not stand still.
- Defective tray: one too deep for the card to be seen in, or too shallow to hold the card flat.
- Unreadable RMA card: one whose numbers cannot be read in the environment camera, or that arrived bent, wet or torn.
- Item too heavy or too large for one gripper to carry level, or with no flat clear face large enough for a grade label. Reset error, not a violation.
- The mailer holding loose fill, a second item or a second card at episode start. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Free one tab of the mailer flap
2. Hold the mailer steady
3. Turn the mailer flap back
4. Take the RMA card out of the mailer
5. Lay the RMA card on the card spot
6. Take the item out of the mailer
7. Lay the item on the inspect mat
8. Read the RMA card
9. Turn the item to show one face
10. Take one grade label from the sheet
11. Hold the item steady on the mat
12. Stick the grade label on the item
13. Lift the item from the inspect mat
14. Place the item in its grade bin
15. Lift the RMA card from the card spot
16. Place the RMA card in the paperwork tray
17. Return both arms home and end the episode